1. Coffee Leaf Rust (*Hemileia vastatrix*)

Sources of Information

- Perfect Daily Grind: Guide to Coffee Pests & Diseases
- ScienceDirect: Coffee Leaf Rust Management

Symptoms

- Yellow-orange powdery **spores** on the underside of leaves.
- Premature **leaf drop**, leading to reduced photosynthesis.
- Severe cases result in **defoliation**, weakened plants, and reduced yields.

Preventive Actions & Why They Work

✅ **Plant Resistant Varieties** (*e.g., Catimor, Sarchimor*)

- **Why?** Some cultivars have **genetic resistance** to leaf rust.

✅ **Prune for Better Airflow**

- **Why?** Reduces humidity, which is necessary for spore germination.

✅ **Balanced Fertilization (Avoid Excess Nitrogen)**

- **Why?** Excess nitrogen promotes **soft leaf growth**, which is more susceptible to infection.

Corrective Actions – Biological/Natural Methods

🌱 **Apply Bordeaux Mixture (Copper Sulfate + Lime)**

- **Why?** Creates a **protective film** that prevents fungal germination.

🌱 **Use Neem Oil or Trichoderma-Based Biocontrols**

- **Why?** Neem **inhibits fungal spore germination**, and Trichoderma **outcompetes leaf rust fungi**.

Corrective Actions – Chemical Treatments

🧪 **Systemic Fungicides (Triazoles, Strobilurins, Copper-based Fungicides)**

- **Mode of Action:**
  - **Triazoles:** Disrupt fungal cell membrane production.
  - **Strobilurins:** Block fungal respiration.
  - **Copper-based fungicides:** Act as a **contact barrier** against spores.

Regenerative Farming Recommendations

1️⃣ **Shade Management:** Maintain diverse **shade trees** to buffer temperature changes that trigger rust outbreaks.

2️⃣ **Compost & Biochar Application:** Enhances **soil microbiology**, which supports plant immunity.
3️⃣ **Intercrop with Nitrogen-Fixing Plants:** Improves plant **nutrient balance** to avoid excessive nitrogen.

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2. Coffee Berry Borer (*Hypothenemus hampei*)

Sources of Information

- Perfect Daily Grind: Coffee Berry Borer Guide
- Journal of Economic Entomology: Biological Control of Coffee Berry Borer

Symptoms

- Small **holes** on coffee cherries.
- Premature **fruit drop**.
- Presence of **black beetles inside berries**.

Preventive Actions & Why They Work

✅ **Timely Harvesting of Ripe & Overripe Cherries**

- **Why?** Reduces borer reproduction inside berries.

✅ **Maintain Field Sanitation (Remove Fallen Cherries)**

- **Why?** Eliminates breeding sites.

✅ **Introduce Parasitic Wasps (E.g., *Phymastichus coffea*)**

- **Why?** Wasps **attack borer larvae**, preventing reproduction.

Corrective Actions – Biological/Natural Methods

🌱 **Beauveria bassiana Fungus Application**

- **Why?** This naturally occurring fungus **infects and kills the beetles**.

🌱 **Trap Cropping with Attractive Pheromone Lures**

- **Why?** Diverts borers away from main crops.

Corrective Actions – Chemical Treatments

💉 **Insecticides (Pyrethroids, Neonicotinoids, Spinosad)**

- **Mode of Action:**
  - **Pyrethroids:** Paralyze the insect nervous system.
  - **Neonicotinoids:** Bind to receptors that cause overactivation.
  - **Spinosad:** Disrupts insect neurotransmitters.

Regenerative Farming Recommendations

1️⃣ **Agroforestry Systems with Mixed Canopy Layers**: Encourage **natural predators** like birds and wasps.

2️⃣ **Mulching with Neem Leaves**: Neem contains **natural insecticidal compounds** that deter borers.

3️⃣ **Use of Beneficial Soil Microbes**: Enhances tree health, making them **less attractive to borers**.

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3. Black Twig Borer (*Xylosandrus compactus*)

Sources of Information

- Wikipedia: Black Twig Borer
- Journal of Applied Entomology: Biological Control of Stem Borers

Symptoms

- Sudden **wilting of twigs and small branches**.
- Entry **holes in stems** with frass (insect waste).

Preventive Actions & Why They Work

✅ **Prune and Destroy Infested Branches Early**

- **Why?** Prevents further spread.

✅ **Encourage Natural Predators (Woodpeckers, Parasitic Wasps)**

- **Why?** These predators **attack borer larvae inside the wood**.

✅ **Apply Tree Wound Protectants After Pruning**

- **Why?** Prevents reinfestation.

Corrective Actions – Biological/Natural Methods

🌱 **Use Fungal Biocontrols (Beauveria bassiana, Metarhizium anisopliae)**

- **Why?** These fungi **infect and kill borers inside the bark**.

🌱 **Apply Neem or Garlic-Based Sprays on Tree Bark**

- **Why?** Natural deterrents reduce female beetle colonization.

Corrective Actions – Chemical Treatments

💉 **Systemic Insecticides (Imidacloprid, Acephate, Pyrethroids)**

- **Mode of Action:**
  - **Imidacloprid:** Disrupts insect nervous systems, leading to paralysis.
  - **Acephate:** Interferes with nerve transmission.

Regenerative Farming Recommendations

1️⃣ **Maintain High Organic Matter in Soil**: Healthier trees **resist infestations better**.

2️⃣ **Incorporate Barrier Trees**: Acts as a **buffer zone** against borer spread.

3️⃣ **Avoid Monoculture (Intercrop with Banana for example)**: Diversifies the ecosystem, making trees **less attractive to borers**.

4. Coffee Leaf Miner (*Leucoptera coffeella*)

Sources of Information

- FAO: Integrated Pest Management for Coffee
- Journal of Applied Entomology: Control of Coffee Leaf Miner

Symptoms

- White, **serpentine mines** on leaves.
- Leaves curl and dry out.
- Severe infestations cause **leaf drop**, reducing photosynthesis.

Preventive Actions & Why They Work

✅ **Encourage Natural Predators (E.g., Parasitic Wasps - *Hymenoptera spp.*)**

- **Why?** Parasitic wasps **lay eggs inside miner larvae**, preventing damage.

✅ **Use Reflective Mulch Under Trees**

- **Why?** Reflects sunlight, disorienting adult moths and reducing egg-laying.

✅ **Avoid Excessive Nitrogen Fertilization**

- **Why?** Too much nitrogen produces **tender leaves**, attracting more miners.

Corrective Actions – Biological/Natural Methods

🌱 **Neem Oil Sprays**

- **Why?** Interferes with leaf miner development and reduces egg-laying.

🌱 **Apply Beauveria bassiana (Entomopathogenic Fungus)**

- **Why?** This fungus **kills the larvae inside the leaves**.

Corrective Actions – Chemical Treatments

💉 **Insect Growth Regulators (IGRs) like Diflubenzuron**

- **Mode of Action:** Prevents larvae from molting, stopping their development.

💉 **Spinosad (Natural Insecticide)**

- **Mode of Action:** Attacks the insect nervous system while being **safe for beneficial insects**.

Regenerative Farming Recommendations

1️⃣ **Increase Tree Diversity in Coffee Farms** – Encourages natural predators like birds and parasitic wasps.
2️⃣ **Enhance Soil Health with Microbial Inoculants** – Stronger trees **resist infestations better**.

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5. Brown Eye Spot (*Cercospora coffeicola*)

Sources of Information

- Perfect Daily Grind: Cercospora in Coffee
- Plant Pathology Journal: Fungal Diseases in Coffee

Symptoms

- **Small brown lesions** with yellow halos on leaves.
- Can affect **both leaves and cherries**.
- Causes **defoliation and premature fruit drop**.

Preventive Actions & Why They Work

✅ **Apply Balanced Fertilization (Especially Potassium and Calcium)**

- **Why?** Strengthens leaf tissue, reducing fungal penetration.

✅ **Reduce Leaf Wetness**

- **Why?** Fungal spores germinate in **moist environments**—improve airflow to dry leaves faster.

✅ **Prune Excessive Vegetation**

- **Why?** Lowers **humidity levels**, reducing infection risk.

Corrective Actions – Biological/Natural Methods

🌱 **Apply Bordeaux Mixture or Copper-Based Sprays**

- **Why?** Prevents fungal spore germination.

🌱 **Use Trichoderma-Based Biocontrols**

- **Why?** Competes with *Cercospora* fungi and **reduces disease severity**.

Corrective Actions – Chemical Treatments

🧪 **Contact Fungicides (Copper Hydroxide, Mancozeb)**

- **Mode of Action:** Prevents fungal spore germination.

💉 **Systemic Fungicides (Azoxystrobin, Propiconazole)**

- **Mode of Action:** Disrupts fungal respiration and growth.

Regenerative Farming Recommendations

1️⃣ **Plant Shade Trees** – Helps **regulate leaf moisture levels**.
2️⃣ **Apply Compost Tea Sprays** – Strengthens plant immunity against fungal infections.
3️⃣ **Use Rock Phosphate as a Soil Amendment** – Provides **natural phosphorus** to strengthen cell walls.

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6. Anthracnose (*Colletotrichum spp.*)

Sources of Information

- ScienceDirect: Anthracnose in Coffee
- Plant Disease Journal: Fungal Management in Coffee

Symptoms

- **Dark, sunken lesions** on leaves, stems, and berries.
- Causes **dieback of branches**.
- Severe cases result in **berry rot**.

Preventive Actions & Why They Work

✅ **Avoid Overhead Irrigation**

- **Why?** Fungal spores spread through **water splashes**.

✅ **Improve Air Circulation in Coffee Fields**

- **Why?** Reduces **humidity**, preventing fungal germination.

✅ **Plant Resistant Coffee Varieties (E.g., Sarchimor)**

- **Why?** Has genetic resistance to anthracnose.

Corrective Actions – Biological/Natural Methods

🌱 **Spray with Fermented Plant Extracts (Garlic, Ginger, Chili Mix)**

- **Why?** Natural antifungal properties **reduce disease severity**.

🌱 **Apply Trichoderma-Based Biopesticides**

- **Why?** Competes with *Colletotrichum* fungi, suppressing disease development.

Corrective Actions – Chemical Treatments

💉 **Systemic Fungicides (Tebuconazole, Difenoconazole)**

- **Mode of Action:** Inhibits fungal enzyme production, stopping growth.

🧪 **Copper-Based Fungicides (Bordeaux Mixture, Copper Oxychloride)**

- **Mode of Action:** Forms a protective barrier, preventing fungal infection.

Regenerative Farming Recommendations

1️⃣ **Boost Soil Microbial Diversity with Mycorrhizal Fungi** – Helps **suppress fungal infections** naturally.
2️⃣ **Use Agroforestry Systems** – Reduces **temperature stress**, making trees more resilient.
3️⃣ **Apply Biochar** – Improves **soil structure and drainage**, reducing fungal spread.

7. Coffee Wilt Disease (*Fusarium xylarioides*)

Sources of Information

- FAO: Coffee Wilt Disease Management
- Plant Pathology Journal: Fusarium Wilt in Coffee

Symptoms

- Leaves **wilt suddenly**, turning yellow and drying out.
- **Dark streaks** under the bark when peeled back.
- Root system **decays**, leading to plant death.

Preventive Actions & Why They Work

✅ **Use Certified, Disease-Free Seedlings**

- **Why?** Fusarium spreads through **infected planting material**.

✅ **Improve Soil Drainage**

- **Why?** Fusarium thrives in **waterlogged soils**.

✅ **Remove Infected Trees Immediately**

- **Why?** Prevents **soil contamination** and spread to healthy plants.

Corrective Actions – Biological/Natural Methods

🌱 **Apply Trichoderma-Based Biocontrols**

- **Why?** Trichoderma fungi **outcompete Fusarium**, suppressing its spread.

🌱 **Use Compost Teas with Beneficial Microbes**

- **Why?** Enhances **soil biodiversity**, reducing disease pressure.

Corrective Actions – Chemical Treatments

💉 **Fungicides (Thiophanate-Methyl, Carbendazim)**

- **Mode of Action:** Inhibits Fusarium **spore formation and root infection**.

Regenerative Farming Recommendations

1️⃣ **Rotate Coffee with Legume Cover Crops** – Reduces Fusarium build-up in soil.
2️⃣ **Add Biochar to Soil** – Enhances microbial competition against Fusarium.
3️⃣ **Use Natural Fungicide Sprays (like neem)** – Acts as a preventive treatment.

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8. Green Scale (*Coccus viridis*)

Sources of Information

- Perfect Daily Grind: Coffee Scale Pests
- Journal of Economic Entomology: Scale Insect Control

Symptoms

- Green, **oval-shaped insects** on leaves and stems.
- Sticky **honeydew** secretion leads to **black sooty mold**.
- Weakens plant by **sucking sap** from leaves.

Preventive Actions & Why They Work

✅ **Encourage Natural Enemies (Ladybugs, Parasitic Wasps)**

- **Why?** These insects **feed on scale pests**.

✅ **Prune to Improve Airflow**

- **Why?** Reduces **scale insect colonization sites**.

✅ **Avoid Overuse of Nitrogen Fertilizers**

- **Why?** Excess nitrogen **promotes soft new growth**, attracting scale insects.

Corrective Actions – Biological/Natural Methods

🌱 **Neem Oil or Soap Sprays**

- **Why?** Dissolves scale insect's **protective wax layer**, causing dehydration.

🌱 **Release Biological Control Agents (Lacewings, Ladybugs, *Metarhizium anisopliae*)**

- **Why?** These predators **feed on scales**, reducing their population naturally.

Corrective Actions – Chemical Treatments

💉 **Insect Growth Regulators (Buprofezin, Pyriproxyfen)**

- **Mode of Action:** Prevents nymphs from **developing into adults**.

💉 **Systemic Insecticides (Imidacloprid, Acetamiprid)**

- **Mode of Action:** Blocks insect **nerve impulses**, leading to death.

Regenerative Farming Recommendations

1️⃣ **Increase Soil Organic Matter** – Improves plant health and **resistance to sap-feeding insects**.

2️⃣ **Encourage Beneficial Insects by Reducing Pesticide Use** – Avoids killing natural predators.

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9. White Stem Borer (*Monochamus leuconotus*)

Sources of Information

- Journal of Tropical Entomology: Stem Borers in Coffee
- FAO: Pest Management for Coffee Borers

Symptoms

- **Girdling of stems**, causing branch dieback.
- Presence of **boreholes with frass (wood dust-like waste)**.
- **Wilting and yellowing leaves**.

Preventive Actions & Why They Work

✅ **Inspect and Remove Infested Branches**

- **Why?** Prevents larvae from **spreading to healthy wood**.

✅ **Encourage Natural Enemies (Woodpeckers, Parasitic Wasps)**

- **Why?** These predators attack **borer larvae inside the wood**.

✅ **Apply Tree Wound Sealants on Pruned Branches**

- **Why?** Reduces borer entry points.

Corrective Actions – Biological/Natural Methods

🌱 **Beauveria bassiana-Based Sprays**

- **Why?** Fungal spores **infect and kill borers inside the bark**.

🌱 **Neem Oil and Garlic-Based Sprays**

- **Why?** Repels adult beetles, reducing egg-laying.

Corrective Actions – Chemical Treatments

💉 **Systemic Insecticides (Chlorpyrifos, Thiamethoxam)**

- **Mode of Action:** Penetrates plant tissues, **killing borers inside the stems**.

Regenerative Farming Recommendations

1️⃣ **Maintain High Organic Matter in Soil** – Stronger trees resist borers better.
2️⃣ **Plant Barrier Trees**– Acts as a **buffer zone** against borers.
3️⃣ **Intercrop Coffee with other trees** – Diversifies the ecosystem, reducing borer outbreaks.

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10. American Leaf Spot (*Mycena citricolor*)

Sources of Information

- Wikifarmer: American Leaf Spot in Coffee
- Plant Disease Journal: Fungal Pathogens in Coffee

Symptoms

- **Circular leaf spots** with a **brown center and yellow halo**.
- Leads to **premature leaf drop**, reducing photosynthesis.
- Severe cases result in **twig dieback**.

Preventive Actions & Why They Work

✅ **Maintain Balanced Tree Canopy (Avoid Excess Shade)**

- **Why?** Reduces **high humidity**, which favors fungal growth.

✅ **Improve Drainage in Coffee Fields**

- **Why?** Prevents **water accumulation**, reducing disease risk.

✅ **Apply Organic Fertilizers to Strengthen Plant Immunity**

- **Why?** Healthier trees **resist fungal infections** better.

Corrective Actions – Biological/Natural Methods

🌱 **Use Trichoderma-Based Biopesticides**

- **Why?** Competes with *Mycena citricolor* fungi, suppressing infection.

🌱 **Spray with Compost Tea or Fermented Plant Extracts**

- **Why?** Boosts leaf immunity and prevents infection spread.

Corrective Actions – Chemical Treatments

💉 **Contact Fungicides (Copper Hydroxide, Mancozeb)**

- **Mode of Action:** Creates a protective barrier, preventing fungal spore germination.

🧪 **Systemic Fungicides (Azoxystrobin, Propiconazole)**

- **Mode of Action:** Disrupts fungal respiration, stopping growth.

Regenerative Farming Recommendations

1️⃣ **Increase Tree Diversity to Reduce Microclimate Humidity** – Reduces fungal outbreaks.
2️⃣ **Boost Soil Health with Mycorrhizal Fungi** – Enhances **disease resistance**.
3️⃣ **Apply Biochar to Improve Soil Aeration** – Limits fungal root infections.